

# Seeing Speech: Magnetic Resonance Imaging-Based Vocal Tract Deformation Visualization Using Cross-Modal Transformer

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## ABSTRACT

As an essential component to advance speech science, understanding of speech production can be greatly helpful to improve our understanding of motor control, dynamical systems of humans during natural speech. Different medical imaging modalities have been leveraged to visualize the dynamic process, in which Magnetic resonance imaging (MRI) provides a valuable tool for evaluating static postures. In this paper, we present our solution to visualize the vocal tract deformation, leveraging the correlation between the MRI and the acoustical signals. We first formulate the problem as a cross-modal prediction task and a novel cross-modal Transformer network is proposed. Thus, we can infer the deformation of the vocal tract by only utilizing the acoustical signals. Then, we present an interactive framework, which can be used to visualize the deformation utilizing the aforementioned network. We hope our solution can also be helpful in pronunciation training for children with sound speech disorders and second language learning.

## CCS CONCEPTS

• Computing methodologies → Image representations; • Networks → Network architectures.

## KEYWORDS

cross-modal generation, audio-visual, vision transformer

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## 1 INTRODUCTION

As a complicated, dynamic behavior of humans, speech production plays a key role in the hearing, perception, and nervous system. During the last decades, different medical imaging modalities and sensory streams have been used to describe the highly variable movements of various articulators, including the lips, jaw, and

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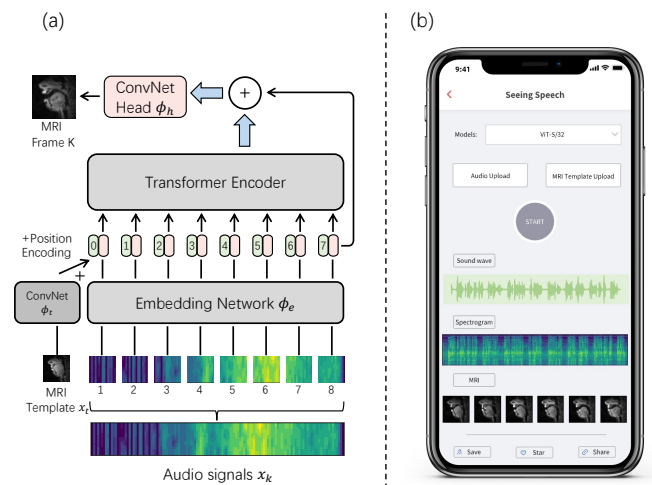
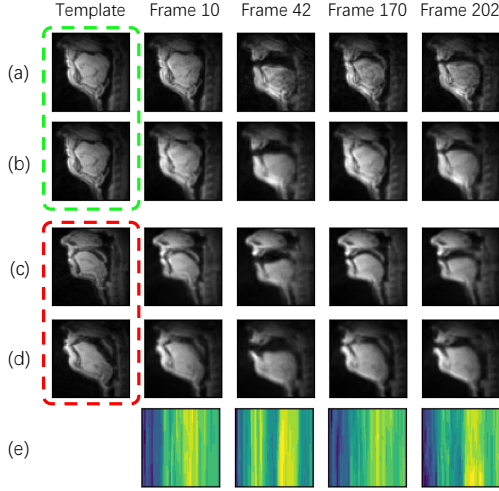


Figure 1: The framework of the proposed approach. (a) presents the cross-modal Transformer Architectures, while the snapshot of the interactive interface is given in (b).

tongue. Modeling of the speech production has many evident applications, including the silent speech interfaces [2], articulatory-based speech synthesis and speech disorder diagnose [10].

However, a long-standing issue in speech production studies concerns the association between the audio signal and the vocal tract configurations [9]. How to estimate the vocal tract configuration from the audio signals and “seeing speech” from the acoustic signals, are still under-explored in previous studies. Building the association between the audio signal and the vocal tract configuration not only can stimulate the understanding of the theoretical quest in speech production studies but also can be helpful for pronunciation training (especially for children with sound speech disorders).

In this paper, we describe our solution to visualize the correlation between the audio signals and the vocal tract shapes, thus we can provide a way to predict the deformation of the vocal tract by using acoustical signals. Specifically, we explored to predict the MRI vocal tract image using the audio signals, using a novel cross-modal Transformer network architectures. Thus, the association between the MRI and audio signals can be built. To the best knowledge of the authors, very few attempts have been made to provide a tool to visualize the causal relationship between the audio and the MRI of the vocal tract. Moreover, extracting meaningful articulatory information from MRI image sequences remains a challenging task, despite various approaches that have been proposed.



**Figure 2: Generated outputs using our cross-modal Transformer network. (a) ground truth. (b) results with target template. (c,d) results with random template. (e) spectrogram. The green box represents the same template, and the red box represents different templates. The red and green boxes indicate that our model can adjust the generated MRI frames according to different templates with the same inputs.**

**Table 1: Performance evaluation with different image similarity metrics. Lower MSE and higher SSIM/CW-SSIM mean better performance.**

| Method         | MSE          | SSIM          | CW-SSIM       |
|----------------|--------------|---------------|---------------|
| 3DCNN          | 60.23        | 0.7582        | 0.8872        |
| ConvLSTM       | 49.24        | 0.7882        | 0.9150        |
| Ours(ViT-S/16) | 48.56        | 0.7922        | 0.9162        |
| Ours(ViT-B/16) | <b>47.23</b> | <b>0.8002</b> | <b>0.9230</b> |

## 2 METHODOLOGY

Figure 1 illustrates an overview of the proposed interactive visualization framework the speech production processing using MRI, which consists of two main stages. In the first stage, a novel cross-modal Transformer network is trained to build the association between the audio signals and MRI sequence. While, the interactive user interfaces are designed in the second stage, leveraging the models trained in the previous stage. When the audio is given, the predicted corresponding MRI images are presented in the interface in near real-time, which can be helpful to visualize the whole tract during speech production.

### 2.1 Offline Training

As shown in Figure 1 (a), our cross-modal Transformer architecture is inspired by ViT [3], and with the following differences. First of all, we replace all fully connected layers (patch flatten module, multi-headed self-attention and MLP header) with convolutional architectures in the ViT to keep the size of the feature maps, which is crucial for generation tasks. Secondly, we employed U-Net [5] as our

embedding network to extract audio features. Thirdly, we introduce additional MRI template branches, which enable the model to obtain different outputs based on different templates.

Specifically, the input audio signals corresponding to MRI frame convert into a sequence of 64-dimension log Mel filterbank features. The results in a spectrogram as input to the embedding network to extract audio features. Meanwhile, we extract the first frame as the template frame of each MRI sequence and feed it into to obtain the MRI features. Finally, the mixed features from audio and MRI are input to the multi-head self-attention (MSA) module and ConvNet header to generate the  $K$ -th MRI frame prediction result.

We also implemented 3-dimensional convolutional neural networks (3DCNN) to predict the vocal tract deformation in MRI videos [8]. Except for 3DCNN, we also implemented ConvLSTM (Convolutional Long Short-Term Memory) architecture [10] in our framework, which is designed for spatial-temporal sequence prediction task. Thus, three different architectures have been explored in our framework, while the cross-modal Transformer architecture performs best in our settings, as demonstrated in the following part.

### 2.2 Quantitative evaluation

To quantitatively evaluate the performance of different architectures, including 3DCNN, ConvLSTM, and our cross-modal Transformer Architectures, we use the hold-out validation strategy [7] to evaluate our proposed method. To measure the similarity between the predicted ones the ground truth, three image similarity measurement metrics are used: the Mean Square Error (MSE), structural similarity index (SSIM) [1] and Complex wavelet structural similarity (CW-SSIM) [6].

To implement our networks, we use PyTorch<sup>1</sup>, and we trained our models until convergence by using the Adam [4] optimizer. The batch size is set as 16. The learning rate is initialized as  $10^{-4}$  and divided by 10 after every 8 epochs. As shown in Table 1 and Fig. 2, all the metrics suggested that our proposed model can provide superior performance.

### 2.3 Online Serving and Demonstration

After the training phase, the model will be uploaded to a server automatically, which is equipped with the NVIDIA GTX 3090Ti GPU. A user interface has been developed and tested using a mobile phone (as shown in Figure 1 (d)), although the interface is in the development stage. In the interface, the user can record the audio. By converting the audio signals to the spectrogram, the trained model can be used to generate the MRI image sequences, which can be visualized in an interactive manner.

## 3 CONCLUSION

Understanding and seeing speech is a long-standing goal for speech science. In this demo, we firstly demonstrate that audio signals are capable of predicting the frames of MRI at the pixel level with satisfactory performance, by using a novel cross-modal transformer architecture. Moreover, an interactive interface is provided to visualize the association between audios and MRI images, which can be employed for second language learning and pronunciation training for children with sound speech disorders.

<sup>1</sup><https://pytorch.org/>

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